

GEN-3 SAFE HELICOPTER

DOCUMENT 00

14-TON / 8-ROTOR SAFETY-CENTERED BASELINE

Central Protected Fixed Safety Capsule Configuration

Eight equally sized 8.0 m rotors • 6.0 MW installed power • Target T/W ≥ 1.5

Main body height 5 m • Capsule rise ≈ 1 m • Overall height ≈ 6 m

Conceptual Design Document • Revision 00-T14-R8-6MW

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PROPRIETARY CONCEPT INFORMATION

Concept / Pre-Feasibility Baseline — Not Certified

Controlled baseline: 14-ton design mass with full fuel, eight 8.0 m rotors, 6.0 MW installed power (6 × 1000 kW class), 2,000 L fuel, overall envelope 24 m × 24 m × 6 m, main body height 5 m, capsule rise ≈ 1 m, fixed central safety capsule, seven external mechanical damper landing legs, six capsule-side isolation dampers, ten-layer safety architecture. Target real thrust-to-weight ratio ≥ 1.5 (real hover lift capability ≈ 21–22 tons).

1. Document Control

Item	Controlled Value
Document Number	00
Revision	00-T14-R8-6MW
Document Title	14-Ton Helicopter 8x8 — Controlled Baseline
Project	Gen-3 Safe Helicopter
Configuration ID	GEN3-SH-14T-R8-6MW
Design mass (full fuel)	14,000 kg
Rotor configuration	8 × 8.0 m equally sized, distributed around central capsule
Central roof rotor	None — no primary lift rotor above occupied capsule
Installed power	6.0 MW (6 × 1000 kW class turboshaft)
Fuel capacity	2,000 L
Overall envelope	24 m × 24 m × 6 m
Main body height	5 m
Capsule rise above body	≈ 1 m
External landing legs	7 mechanical damper legs
Capsule isolation dampers	6 mechanical damper elements
Safety capsule	Fixed / non-jettisonable / centrally protected
Target T/W	≥ 1.5 (real lift ≈ 21–22 t)
Founder	Amirbahador Atai
Status	Concept / pre-feasibility; not certified

This document is a design baseline, not a type certificate, flight manual, structural substantiation package or verified performance record.

2. Executive Summary

This Document 00 establishes the controlled baseline for the Gen-3 Safe Helicopter in the 14-ton class. The configuration is sized to achieve a real thrust-to-weight ratio of at least 1.5 at the full-fuel design mass of 14,000 kg, requiring a real hover lift capability of approximately 21–22 tons.

- Design mass with full fuel: **14,000 kg**.
- Eight primary lift rotors, each **8.0 m** diameter, distributed around the central capsule; no rotor directly above the occupied volume.
- Total geometric disk area $\approx$ **402.1 m²**; disk loading at 14 t $\approx$ 34.8 kg/m².
- Installed power: **6.0 MW** (six 1000 kW-class turboshaft prime movers).
- Approximate real hover lift capability: **21–22 tons** → target T/W $\geq$ 1.5.
- Fuel capacity: **2,000 L** ($\approx$ 1,600 kg at 0.80 kg/L).
- Overall envelope: **24 m × 24 m × 6 m**.
- Main body height: **5 m**; capsule rise above main body: $\approx$ **1 m**.
- Central fixed safety capsule with multi-angle carbon-fiber roll-cage, high-stroke isolation and soft damper-mounted seats.
- Seven external mechanical damper landing legs + six capsule-side isolation damper elements.
- Full ten-layer safety architecture retained.
- Aircraft-level emergency recovery concept (8 advanced parachutes).

The central engineering proposition remains layered survivability: structural containment, mechanical isolation, multiple damped load paths, fire protection, distributed propulsion, fault reconfiguration, emergency recovery and multi-stage occupant energy management.

3. Frozen / Controlled Product Baseline

Parameter	Baseline Value	Status
Design mass (full fuel)	14,000 kg	Controlled target
Rotor count	8	Controlled
Rotor diameter	8.0 m (all equal)	Controlled
Rotor arrangement	Distributed around central capsule; no central-roof rotor	Controlled
Total geometric disk area	≈ 402.1 m²	Calculated
Disk loading at 14 t	≈ 34.8 kg/m²	Calculated
Installed power	6.0 MW	Controlled
Prime movers	6 × 1000 kW-class turboshaft	Controlled direction
Approx. real hover lift capability	21 – 22 tons	Calculated [C-A]
Target thrust-to-weight ratio	≥ 1.5 at 14 t	Controlled objective
Fuel capacity	2,000 L	Controlled
Fuel mass (reference)	≈ 1,600 kg at 0.80 kg/L	Planning
Overall length	24 m	Controlled target
Overall width	24 m	Controlled target
Overall height	6 m	Controlled target
Main body height	5 m	Controlled
Capsule rise above main body	≈ 1 m	Controlled
External landing legs	7 mechanical damper legs	Controlled
Capsule-side dampers	6 mechanical damper elements	Controlled
Safety capsule	Fixed / non-jettisonable / centrally protected	Controlled
Occupant capacity	7	Controlled
Safety architecture	Ten-layer concept	Controlled
Recovery concept	8-parachute aircraft-level concept	Controlled

4. Vehicle Architecture

The platform is organized around a centrally located, protected safety capsule. Eight equally sized 8.0 m rotors are distributed around the capsule so that no primary lift rotor is positioned directly above the occupant volume. Overall envelope is set at 24 m × 24 m × 6 m to provide comfortable tip clearance between rotors and adequate ground clearance.

Main body height is controlled at 5 m. The fixed safety capsule is positioned in the central heart of the vehicle and rises approximately 1 m above the main body, giving the nominal overall height of 6 m.

- Central occupant capsule near vehicle CG, raised ≈ 1 m above the 5 m main body.
- Primary airframe forms a load-carrying perimeter around the capsule.
- Eight rotor stations distributed for lift redundancy and control authority.
- Six 1000 kW-class turboshaft prime movers feeding a segmented electrical distribution system to eight independent rotor channels.
- Fuel and hot machinery separated from the occupied capsule by dedicated containment zones.
- Seven external mechanical damper landing legs + six capsule-side isolation damper elements.
- Architecture prioritizes controlled graceful degradation.

5. Ten-Layer Safety Architecture

Layer	Function	Implementation Concept
1	Occupant Survival Capsule	Central fixed capsule with thick multi-angle carbon-fiber roll-cage
2	Multi-stage Energy Absorption	Landing-leg response → airframe deformation → capsule-mount stroke → roll-cage → soft seats
3	Capsule-to-Airframe Isolation	High-stroke mechanical damper mounts + 6 capsule-side damper elements
4	Distributed Propulsion	8 independent rotor channels
5	Prime-Mover Redundancy	6 independent 1000 kW-class turboshaft units
6	Fire Detection / Isolation / Suppression	Zone detection, isolation, suppression around capsule and critical zones
7	Control Redundancy	Segregated sensing, computation and actuator paths
8	Emergency Descent / Recovery	8 advanced parachutes + degraded flight modes
9	Water Survival	Buoyancy / flotation concept, multi-orientation egress
10	Structural Containment & Modular Isolation	High-energy hazards outside occupant volume; modular isolation of failed/burning rotor

Safety claims remain conceptual until supported by analysis and test. No under-capsule fans in this fixed-capsule baseline.

6. Fixed Safety Capsule – Detailed Features

Feature	Specification
Location	Central heart of vehicle; rises ≈ 1 m above 5 m main body
Overall height contribution	Main body 5 m + capsule rise ≈ 1 m → overall 6 m
Primary volume shape	Near-spherical / highly rounded
Internal reinforcement	Thick multi-angle carbon-fiber roll-cage
Occupant seats	7 soft bed-like seats with high-stroke mechanical dampers
Capsule-to-airframe interface	Multiple high-stroke mechanical damper mounts
Capsule-side dampers	6 mechanical damper elements (separate from landing legs)
External landing legs	7 mechanical damper legs with progressive energy absorption
Fire protection	Suppression nozzles at capsule perimeter
Egress	Multiple exit routes; design objective to remain operable after power loss
Parachute system	8 advanced canopies (aircraft-level recovery concept)
Independent capsule flight	None — fixed non-jettisonable

7. Eight-Rotor Propulsion Architecture (8.0 m)

Item	Value
Rotor count	8 (all equal)
Rotor diameter	8.0 m
Arrangement	Distributed around central capsule; no central-roof rotor
Area per rotor	50.27 m²
Total geometric disk area	≈ 402.1 m²
Disk loading at 14,000 kg	≈ 34.8 kg/m²
Hover thrust required at 14 t	≈ 137.3 kN total (≈ 17.2 kN per rotor)
Ideal induced power at 14 t (ISA SL)	≈ 1.82 MW
Approx. real power at 14 t (FM ≈ 0.72)	≈ 2.53 MW
Approx. real power for 21 t lift (FM ≈ 0.72)	≈ 4.65 MW
Approx. real power for 22 t lift (FM ≈ 0.72)	≈ 5.00 MW
Installed power	6.0 MW
Approx. real hover lift capability	21 – 22 tons
Resulting T/W at 14 t	≈ 1.50 – 1.57

Installed power of 6.0 MW provides margin above the approximate real power required for 21–22 t lift, supporting the target $T/W \geq 1.5$ after realistic losses and reserves.

8. Propulsion and Engine Architecture

Item	Value
Number of prime movers	6
Power class per unit	1000 kW class
Total installed power	6.0 MW
Power flow concept	Turboshaft → generator → segmented HV bus → independent inverters → rotor motors
Redundancy objective	Continue controlled flight/landing after loss of any single prime mover or any single rotor channel
Key safety rule	Physical independence of channels; shared vulnerable buses do not equal independent redundancy

Final OEM selection, generator matching and installation derate remain open. Published engine figures are engine-level only.

9. Fuel System, Endurance and Range

Parameter	Value
Nominal tank capacity	2,000 L
Reference fuel mass	≈ 1,600 kg (0.80 kg/L)
Approx. real hover power at 14 t	≈ 2.53 MW
Hover endurance order-of-magnitude at 14 t	Approximately 90–120 minutes (typical SFC)
Cruise endurance order-of-magnitude	Approximately 3.0–4.0 hours
Range order-of-magnitude at 200 km/h	Approximately 600–800 km (with reserves)

Endurance and range are order-of-magnitude estimates only. Not certified performance.

10. Preliminary Mass Budget

Mass group	Planning mass (kg)	Comment
Primary airframe & rotor support	2,600 – 3,000	Larger 8×8 m system
Safety capsule & protection	1,150 – 1,350	Roll-cage, isolators, seats
Eight rotors, hubs & control (8.0 m)	1,400 – 1,700	Supplier-dependent
Powerplant / generators / drives (6 MW)	2,400 – 2,800	Six 1000 kW-class units
Fuel system (dry)	300 – 360	Tanks, pumps, protection
Fuel (2,000 L)	1,600	0.80 kg/L
Avionics, flight control, electrical	650 – 750	Redundancy included
Landing gear (7+6 dampers)	500 – 600	Mechanical damper legs
Fire, recovery, safety equipment	600 – 700	8-parachute concept
Thermal management	500 – 600	Higher power level
Interconnects, fluids, growth reserve	1,100 – 1,400	Margin
Total planning mass (full fuel)	≈ 13,000 – 14,800	Design target 14,000 kg

Mass status: OPEN. 14,000 kg is the controlled design target with full fuel, not a closed weight statement.

11. Flight-Control and Fault-Reconfiguration

The eight-rotor distributed layout with six independent prime movers supports graceful degradation. The flight-control system must detect abnormal rotor, power and actuator behavior and reconfigure within validated limits. Independent monitoring of each rotor channel, automatic isolation of a failed channel, and reallocation of authority across healthy rotors are design objectives.

12. Structural and Load-Path Philosophy

Multiple load paths around the central capsule are retained. Seven external mechanical damper legs and six capsule-side damper elements form the primary ground-impact and isolation load paths. Main body height is 5 m; capsule rises ≈ 1 m. Structural sizing will use limit and ultimate load cases. No certification compliance is claimed.

13. Emergency Recovery Concept

Scenario	Primary response	Secondary response
Single rotor failure	Control reconfiguration	Power redistribution / landing
Multiple rotor degradation	Emergency flight mode	8-parachute recovery
Major fire	Detection, isolation, power reduction	Emergency recovery / landing
Severe structural event	Maintain capsule integrity	Aircraft recovery concept
Water impact / ditching	Capsule survivability and egress	Flotation provisions

14. Fire, Thermal and Water Survival

Fire: zone detection, fuel/electrical isolation, barriers, suppression around capsule and critical zones, load shedding. Thermal management mass is increased for the 6 MW power level. Water: capsule intended to provide protected survival volume, controlled flotation concept and multi-orientation egress. No claim is made that the airframe remains intact in all water-impact conditions.

15. Manufacturing and Prototyping Strategy

Phase 1: material and coupon tests. Phase 2: subscale rotor/motor/controller tests. Phase 3: full-scale capsule and damper impact articles. Phase 4: integrated 6 MW-class propulsion test rig. Phase 5: unmanned demonstrator. Phase 6: progressive flight-envelope expansion. Phase 7: certification-oriented programme.

16. Verification and Validation Plan

Subsystem	Analysis	Bench / Dynamic	Flight
Rotor / hub (8.0 m)	Yes	Yes	Yes
Propulsion (6 MW)	Yes	Yes	Yes
Flight controls	Yes	HIL + fault injection	Yes
Capsule structure	FEA	Full-scale impact	Limited
Dampers (7+6)	Model + test	Full-scale	Integrated
Fire protection	Hazard analysis	System test	As required
Emergency recovery	Simulation	Deployment test	Flight demo
Water survival	Simulation	Water-impact test	As required

17. Certification and Safety Assessment

Certification pathway is not fixed. Applicable basis depends on final category, MTOM, occupant capacity and jurisdiction. This document does not claim compliance. Required safety work includes FHA, PSSA/SSA, common-cause analysis, structural and load-path assessment, human-factors and egress assessment, and full verification traceability. The ten-layer architecture is not ten independent guarantees.

18. Product and IP Positioning

Differentiating theme: integrated architecture combining centrally protected fixed capsule, structural isolation, multi-stage energy absorption, distributed 8x8.0 m propulsion (no rotor above capsule), 6 MW power for T/W ≥ 1.5, fire containment and emergency recovery as one safety system. Assets remain candidate IP / know-how until formal legal assessment.

19. Preliminary Risk Register

Risk	Impact	Treatment	Next action
Mass growth above 14 t	High	14 t target + reserve	Supplier data + weighed hardware
Real lift below 21 t	High	6.0 MW installed	Rotor CFD + efficiency model
Rotor mass higher than planned	High	Planning range	Supplier blade/hub data
Control with 8 large rotors	Medium-High	Distributed layout	Control-law sim + HIL
Damper stroke insufficient	High	7+6 elements	Dynamic impact test
Thermal management shortfall	High	Increased cooling mass	Thermal analysis
Common-mode failures	High	Independent channels	SSA + HIL
Envelope / clearance issues	Medium	24x24x6 m target	Detailed packaging study

20. Open Engineering Items

- Freeze detailed mission profile.
- Complete 8.0 m rotor aerodynamic model (blade count, solidity, tip speed).
- Final OEM selection and generator matching for 6 × 1000 kW class.
- Close full power budget including climb and degraded-mode margins.
- Replace planning mass with component-level data; close mass at 14 t.
- Capsule structural load cases and damper stroke/force targets.
- Parachute sizing, deployment envelope and attachment loads.
- Water-impact and flotation requirements.
- Formal system safety architecture and failure-mode database.
- Map to applicable certification basis.
- Confirm 24×24×6 m envelope after detailed rotor clearance study.

21. Design Rules

- Rule 1** — 14 t full-fuel mass is a controlled target, not paper-only.
- Rule 2** — Do not claim safety credit before evidence exists.
- Rule 3** — Prefer passive mechanical safety layers where practical.
- Rule 4** — Avoid common-mode dependencies that defeat multiple layers.
- Rule 5** — Treat the capsule as a system, not only a stronger cabin.
- Rule 6** — Maintain $T/W \geq 1.5$ as a controlled design driver.
- Rule 7** — Separate controlled baseline, targets, calculations and demonstrated performance.
- Rule 8** — Main body 5 m and capsule rise ≈ 1 m are controlled geometric targets.

22. Preliminary Performance Envelope

Metric	Value
Design mass (full fuel)	14,000 kg
Total disk area	≈ 402.1 m²
Disk loading at 14 t	≈ 34.8 kg/m²
Ideal induced power at 14 t	≈ 1.82 MW
Approx. real power at 14 t (FM ≈ 0.72)	≈ 2.53 MW
Approx. real power for 21 t lift	≈ 4.65 MW
Approx. real power for 22 t lift	≈ 5.00 MW
Installed power	6.0 MW
Approx. real hover lift capability	21 – 22 tons
Target / approx. T/W at 14 t	≥ 1.5 (≈ 1.50 – 1.57)
Fuel capacity	2,000 L (≈ 1,600 kg)
Overall envelope	24 m × 24 m × 6 m
Main body height	5 m
Capsule rise	≈ 1 m
Hover endurance (order-of-magnitude)	90–120 min at 14 t
Cruise endurance (order-of-magnitude)	3.0–4.0 hours
Range (order-of-magnitude)	600–800 km at 200 km/h

All performance figures are preliminary estimates. Not certified performance.

23. Investor and Due-Diligence Notes

This baseline is explicitly sized for real $T/W \geq 1.5$ at 14 t full-fuel mass using eight 8.0 m rotors and 6.0 MW installed power. Main body height 5 m and capsule rise ≈ 1 m define the 6 m overall height. Safety architecture, fixed capsule, 7+6 damper system and recovery concept are retained. Due-diligence should focus on mass closure at 14 t, rotor performance of the 8.0 m rotors, thermal management of the 6 MW package, and physical demonstration of energy-absorption claims.

24. Glossary

Term	Meaning
T/W	Thrust-to-weight ratio (real lift capability / weight)
FM	Figure of Merit
Installed power	Total power available at aircraft level after installation losses
Fixed capsule	Non-jettisonable centrally protected occupant capsule
Main body height	5 m controlled
Capsule rise	≈ 1 m above main body
Overall height	6 m (5 m body + ≈ 1 m capsule rise)

25. Final Baseline Statement

Document 00 (Revision 00-T14-R8-6MW), titled **14-Ton Helicopter 8x8**, establishes the controlled Gen-3 baseline: design mass **14,000 kg** with full fuel; eight equally sized **8.0 m** rotors distributed around a centrally protected fixed safety capsule (no primary lift rotor above the capsule); installed power **6.0 MW** (six 1000 kW-class turboshaft units); approximate real hover lift capability **21–22 tons** supporting target thrust-to-weight ratio ≥ 1.5 ; fuel capacity **2,000 L**; overall envelope **24 m × 24 m × 6 m**; main body height **5 m**; capsule rise $\approx 1\text{ m}$; seven external mechanical damper landing legs; six capsule-side isolation damper elements; and full ten-layer safety architecture.

The baseline remains deliberately conservative about claims. Quantitative performance, mass closure, power installation and certification compliance remain subject to analysis and test. Next revision should close detailed rotor aerodynamics, OEM engine package, component-level mass statement at 14 t, and envelope confirmation after clearance studies.

Founder / Principal Author: **Amirbahador Atai**

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END OF DOCUMENT 00 — 14-TON / 8x8.0 m / 6 MW / T/W ≥ 1.5